Series of real numbers

1. Series with real terms

Def. A series $\sum_{n=1}^{\infty} a_n$ is called absolutely convergent if the series $\sum_{n=1}^{\infty} |a_n|$ is convergent.

Th 1. Every absolutely convergent series is convergent.

Def. A convergent series $\sum_{n=1}^{\infty} a_n$ which is not absolutely convergent is called a semiconvergent series.

ex. $\sum_{n=1}^{\infty} \frac{(-1)^{n-1}}{n} = \ln 2$ is convergent, but $\sum_{n=1}^{\infty} \left| \frac{(-1)^{n-1}}{n} \right| = \sum_{n=1}^{\infty} \frac{1}{n}$ is divergent. Hence it is a semiconv. seri

Th 2. (Abel - Dirichlet test) Let (a_n), $n \geq 1$ be a monotonically decreasing sequence with $\lim a_n = 0$ and (b_n), $n \geq 1$ a sequence such that the series $\sum_{n=1}^{\infty} b_n$ has a bounded sequence of partial sums. Then the series $\sum_{n=1}^{\infty} a_n \cdot b_n$ is convergent.

Ex. $\sum_{n=1}^{\infty} \frac{\sin n}{\sqrt{n}}$, take $a_n = \frac{1}{\sqrt{n}} \searrow 0$ and $b_n = \sin n$

$$S_n = b_1 + b_2 + \ldots + b_n = \sin 1 + \sin 2 + \ldots + \sin n$$

$$\sin x + \sin 2x + \sin 3x + \ldots + \sin nx = \frac{\sin \frac{nx}{2} \cdot \sin \frac{(n+1)x}{2}}{\sin \frac{x}{2}}$$

$x \in \mathbb{R} \setminus \{2k\pi \mid k \in \mathbb{Z}\}$

Let $x = 1 \Rightarrow S_n = \sin 1 + \dots + \sin n = \dfrac{\sin\frac{n}{2}\sin\frac{(n+1)}{2}}{\sin\frac{1}{2}}$

$\in \left[-\dfrac{1}{\sin\frac{1}{2}}, \dfrac{1}{\sin\frac{1}{2}}\right] \Rightarrow \sum_{n=1}^{\infty} \dfrac{\sin n}{\sqrt{n}}$ is conv.

Def. A series of the form $a_1 - a_2 + a_3 - a_4 + \dots = \sum_{n=1}^{\infty} (-1)^{n-1} a_n$, $a_n > 0$, $\forall n \geq 1$ is called an alternating series.

Th 3. (Leibniz test) Let (a_n), $n \geq 1$ be a decreasing sequence with $\lim a_n = 0$. Then the alternating series:

$$\sum_{n=1}^{\infty} (-1)^{n-1} a_n \text{ is convergent.}$$

2. Properties of series:

Th 1. Let $\sum_{n=1}^{\infty} a_n$ be an absolutely convergent series with sum s. Then for every bijection $\sigma: \mathbb{N}^* \to \mathbb{N}^*$ the series $\sum_{n=1}^{\infty} a_{\sigma(n)}$ is absolutely convergent and its sum is also s. (Permuting terms in an abs. conv. series doesn't change either sum or nature)

Th 2. Let $\sum_{n=1}^{\infty} a_n$ be a series with positive terms. Then for every bijection $\sigma: \mathbb{N}^* \to \mathbb{N}^*$ we have.

$$\sum_{n=1}^{\infty} a_n = \sum_{n=1}^{\infty} a_{\sigma(n)}$$

Th 3. Let $\sum_{n=1}^{\infty} a_n$ be a semiconv. series. Then for

every $\alpha \in \overline{\mathbb{R}}$ there exists a permutation $\sigma: \mathbb{N}^* \to \mathbb{N}^*$ s.t.:

$$\sum_{n=1}^{\infty} a_{\sigma(n)} = \alpha \quad \text{(Riemann)}$$

ex. $\sum_{n=1}^{\infty} \frac{(-1)^{n-1}}{n} = 1 - \frac{1}{2} + \frac{1}{3} - \frac{1}{4} + \dots = \ln 2$

Rearrange terms $\to 1 - \frac{1}{2} - \frac{1}{4} + \frac{1}{3} - \frac{1}{6} - \frac{1}{8} + \frac{1}{5} - \frac{1}{10} - \frac{1}{12} + \dots$

$$\alpha_{3n} = \sum_{k=1}^{n} \left(\frac{1}{2k-1} - \frac{1}{4k-2} - \frac{1}{4k} \right) = \frac{1}{2} \sum_{k=1}^{n} \left(\frac{1}{2k-1} - \frac{1}{2k} \right)$$

$$= \frac{1}{2}\left(H_{2n} - H_n \right) = \frac{1}{2} \gamma_n - \frac{1}{2} \gamma_n + \frac{1}{2} \ln 2$$

$$\Rightarrow \lim \alpha_{3n} = \frac{\ln 2}{2} \neq \ln 2$$

Def. Let $\sum_{n=0}^{\infty} a_n$, $\sum_{n=0}^{\infty}$ be two series of real numbers. The series $\sum_{n=0}^{\infty} c_n$, where

$$c_n = \sum_{k=0}^{n} a_k b_{n-k} = a_0 b_n + a_1 b_{n-1} + \dots + a_n b_0 \text{ in}$$

called the **Cauchy Product** of the given series.

Obs. In general, the Cauchy product of two conv. series is not necessarily a convergent series.

Th 4. Let $\sum_{n=0}^{\infty} a_n$ be an absolutely conv. series and

$\sum_{n=0}^{\infty}$

Sequences and series of functions

Example of seq. of functions:

$1, x, x^2, \dots$; $f_n(x) = x^n$, if $n \geq 1$, $(f_n)_{n \geq 0}$, $x \in \mathbb{R}$

$\sin x, \sin 2x, \dots$

Def. A sequence of functions (f_n), $f_n : A \to \mathbb{R}$ is said to be conv. or pointwise conv. to the function $f : A \to \mathbb{R}$ on A if:

$$f(x) = \lim_{n \to \infty} f_n(x)$$

Denote: $f_n \to f$ or $f_n \xrightarrow{PC} f$ the pointwise conv.

ex. $f_n : \mathbb{R} \to \mathbb{R}$

$$f_n(x) = \frac{x + x^2 e^{nx}}{1 + e^{nx}}, \quad x \in \mathbb{R}, \ n \in \mathbb{N}^*$$

$$\lim_{n \to \infty} f_n(x) = \begin{cases} x, & x < 0 \\ 0, & x = 0 \\ x^2, & x > 0 \end{cases}$$

$\Rightarrow f_n \xrightarrow{PC} f$, where f is that relation

Th 1. (A test of uniform convergence) Let $(f_n)_{n \geq 1}$, $f_n : A \to \mathbb{R}$ be a seq. of functions and $f : A \to \mathbb{R}$. If $\exists$ $(a_n)_{n \geq 1}$, $a_n > 0$ s.t.:

1. $\lim a_n = 0$
2. $|f_n(x) - f(x)| \leq a_n$, for all $n \geq n_0$ and all $x \in A$

$\Rightarrow f_n \xrightarrow{UC} f$ on A.

ex. $f_n: \mathbb{R} \to \mathbb{R}$, $f_n(x) = \frac{\cos x + 2n \sin x}{n+1}$

$\lim f_n = 2\sin x$, $x \in \mathbb{R}$

$$|f_n - f| = \left| \frac{\cos x + 2n\sin x}{n+1} - 2\sin x \right|$$

$$= \left| \frac{\cos x + 2n\sin x - 2n\sin x - 2\sin x}{n+1} \right| = \left| \frac{\cos x - 2\sin x}{n+1} \right| \leq$$

$\frac{3}{n+1} \Rightarrow |f_n(x) - f(x)| \leq \frac{3}{n+1}$, $\forall n \in \mathbb{N}^*$, $\forall x \in \mathbb{R}$

$a_n = \frac{3}{n+1} \xrightarrow[n\to\infty]{} 0 \Rightarrow f_n \xrightarrow{UC} f$ on $\mathbb{R}$

Th 2. Let (f_n), $n \geq 1$, $f_n: A \to \mathbb{R}$, $A \subseteq \mathbb{R}$ be a seq. of continous functions, $f_n \xrightarrow{UC} f$ on $A \Rightarrow f$ is continous on A. (Preservation of continuity)

Th 3. Let (f_n), $n \geq 1$, $f_n: [a,b] \to \mathbb{R}$ be a seq of continous functions, $f_n \xrightarrow{UC} f$ on $[a,b] \Rightarrow$

$$\lim_{n\to\infty} \int_a^b f_n(x)\,dx = \int_a^b f(x)\,dx$$

Th 4. Let (f_n), $n \geq 1$, $f_n: [a,b] \to \mathbb{R}$ be a seq of differentiable functions s.t.:

1. $f_n \xrightarrow{PC} f$ on $[a,b]$
2. $f_n' \xrightarrow{UC} f'$ on $[a,b]$

$\Rightarrow f$ is differentiable on $[a,b]$

$$\Rightarrow \left(\lim_{n\to\infty} f_n\right)' = \lim_{n\to\infty} f_n'$$

Def. The series of functions $\sum_{n=1}^{\infty} f_n$ is said to be pointwise (uniformly) conv. if the seq of partial sums

$(f_n)_{,\, n\geq 1}$ is pointwise (uniformly) conv.

Th 5. (Weierstrass) Let $\sum_{n=1}^{\infty} f_n$, $f_n : A \to \mathbb{R}$ be a series of functions and $(a_n)_{,\, n\geq 1}$, $a_n > 0$ s.t.:

1. $|f_n(x)| \leq a_n$, $\forall n \geq n_0$, $x \in A$
2. $\sum_{n=1}^{\infty} a_n$ is conv.

Then $\sum_{n=1}^{\infty} f_n$ is uniformly and absolutely conv. on A

ex. $\sum_{n=1}^{\infty} \frac{\cos nx}{n^2+x^2}$, $f_n(x) = \frac{\cos nx}{n^2+x^2}$

$|f_n(x)| \leq \frac{1}{n^2+x^2} \leq \frac{1}{n^2}$ $\Rightarrow \sum_{n=1}^{\infty} f_n$ is un. and abs. conv.

$\sum_{n=1}^{\infty} \frac{1}{n^2}$ is conv.